

Combining Radiomics and Autoencoders to Distinguish Benign and Malignant Breast Tumors on US Images

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Background: US is clinically established for breast imaging, but its diagnostic performance depends on operator experience. Computer-assisted (real-time) image analysis may help in overcoming this limitation.

Purpose: To develop precise real-time-capable US-based breast tumor categorization by combining classic radiomics and autoencoder-based features from automatically localized lesions.

Materials and Methods: A total of 1619 B-mode US images of breast tumors were retrospectively analyzed between April 2018 and January 2024. nnU-Net was trained for lesion segmentation. Features were extracted from tumor segments, bounding boxes, and whole images using either classic radiomics, autoencoder, or both. Feature selection was performed to generate radiomics signatures, which were used to train machine learning algorithms for tumor categorization. Models were evaluated using the area under the receiver operating characteristic curve (AUC), sensitivity, and specificity and were statistically compared with histopathologically or follow-up–confirmed diagnosis.

Results: The model was developed on 1191 (mean age, 61 years $\pm$ 14 [SD]) female patients and externally validated on 50 (mean age, 55 years $\pm$ 15]). The development data set was divided into two parts: testing and training lesion segmentation (419 and 179 examinations) and lesion categorization (503 and 90 examinations). nnU-Net demonstrated precision and reproducibility in lesion segmentation in test set of data set 1 (median Dice score [DS]: 0.90 [IQR, 0.84–0.93]; $P = .01$) and data set 2 (median DS: 0.89 [IQR, 0.80–0.92]; $P = .001$). The best model, trained with 23 mixed features from tumor bounding boxes, achieved an AUC of 0.90 (95% CI: 0.83, 0.97), sensitivity of 81% (46 of 57; 95% CI: 70, 91), and specificity of 87% (39 of 45; 95% CI: 77, 87). No evidence of difference was found between model and human readers (AUC = 0.90 [95% CI: 0.83, 0.97] vs 0.83 [95% CI: 0.76, 0.90]; $P = .55$ and 0.90 vs 0.82 [95% CI: 0.75, 0.90]; $P = .45$) in tumor classification or between model and histopathologically or follow-up–confirmed diagnosis (AUC = 0.90 [95% CI: 0.83, 0.97] vs 1.00 [95% CI: 1.00, 1.00]; $P = .10$).

Conclusion: Precise real-time US-based breast tumor categorization was developed by mixing classic radiomics and autoencoder-based features from tumor bounding boxes.

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US imaging is routinely applied to detect and characterize breast lesions (1,2). It is broadly available and facilitates real-time tumor diagnosis at high image resolution. However, its diagnostic value strongly depends on the experience and skill of the operator (3,4). Although modern picture archiving and communication systems, or PACS, enable storage of US cine clips for comprehensive re-evaluation, the real-time lesion analysis during US examinations remains irreplaceable. The development of computer-aided (real-time) image analysis can support physicians during and after examinations and can help to overcome the limitations of breast US (4).

Radiomics, a process assisting the conversion of imaging data into quantifiable features, has the potential to

facilitate US image–based breast cancer diagnosis (5,6). Classically, radiomics is performed by manual lesion segmentation and extraction of features using predefined statistical image descriptors (7,8). Alternatively, image-based features can be calculated without user definition, for example, on labeled data using convolutional neural networks or without annotations using autoencoders. All methods help extract traits (ie, data points) that reflect the relationship between gray-scale levels of the tumor and its surroundings at the different levels of image abstraction. These interdependencies are expressed, for example, by changing or repeating intensities (4,7).

Commonly, US-based categorization models are developed with clinical features (eg, patient's age or tumor

Abbreviations

AUC = area under the receiver operating characteristic curve,
BI-RADS = Breast Imaging Reporting and Data System,
DS = Dice score

Summary

Combining classical radiomics and autoencoder-based features extracted from tumor bounding boxes improves US-based categorization of benign and malignant breast tumors compared with segmentation-based approaches.

Key Results

- Using retrospective data from 1241 patients with 1232 breast lesions, a US-based model was developed by mixing classic radiomics and autoencoder features.
- The model trained with 23 tumor bounding box–derived features achieved an area under the receiver operating characteristic curve (AUC) of 0.90, sensitivity of 81% (46 of 57), and specificity of 87% (39 of 45).
- No evidence of difference was found between model and human readers (AUC = 0.90 vs 0.83 [$P = .55$] and 0.90 vs 0.82 [$P = .45$]) in breast tumor classification or between model and histopathologically or follow-up–confirmed diagnosis (AUC = 0.90 vs 1.00; $P = .10$).

size) or classic image descriptors extracted from segments manually delineated by experienced radiologists after images were recorded (6,9). It has been reported that computing image-based features not only from the tumor region but also tumor margin may increase the area under the receiver operating characteristics curve (AUC) of breast tumor categorization in MRI (10), US shear-wave elastography (11), and B-mode US with and without Doppler imaging (12,13). Recently, US-based breast cancer diagnosis was performed by sequentially applying different categorization functions (14,15). The assignment of

a breast lesion to its correct phenotype resulted from merging the decisions proposed by multiple independent categorization models trained on classic or deep learning–based features. However, the direct mixing of various types of image-based features for breast tumor categorization on US images has not yet been reported.

Therefore, we investigated whether integrating classic and deep learning–based features improves sensitivity, specificity, and AUC of US-based radiomics for categorization of benign and malignant breast tumors. Furthermore, we evaluated whether extracting features from regions of interest types (ie, bounding boxes) other than segmentation can lead to an effective and potentially real-time–capable computer-aided US-based breast tumor diagnosis system.

Materials and Methods

Local institutional review board approval was obtained for this retrospective study, and informed consent was waived. Part of the study database (497 of 1232 patients) was used in our prior study (16). The generated codes are available online (<https://github.com/ExMILab/USBreastLocalizationClassification>) (Appendix S1).

Study Database: Acquisition, Cleaning, and Splitting

Data for this study were collected between April 2018 and December 2021 and between August 2023 and January 2024 (Fig 1). B-mode US images of histopathologically confirmed (core biopsy) malignant breast tumors and 2-year follow-up or histopathologically confirmed benign breast tumors were included. Exclusion criteria comprised examinations with visible calipers, duplicates, reference mask discrepancies, nonbreast structures, and visible biopsy needles. Image pre-processing was performed on the collected US images that

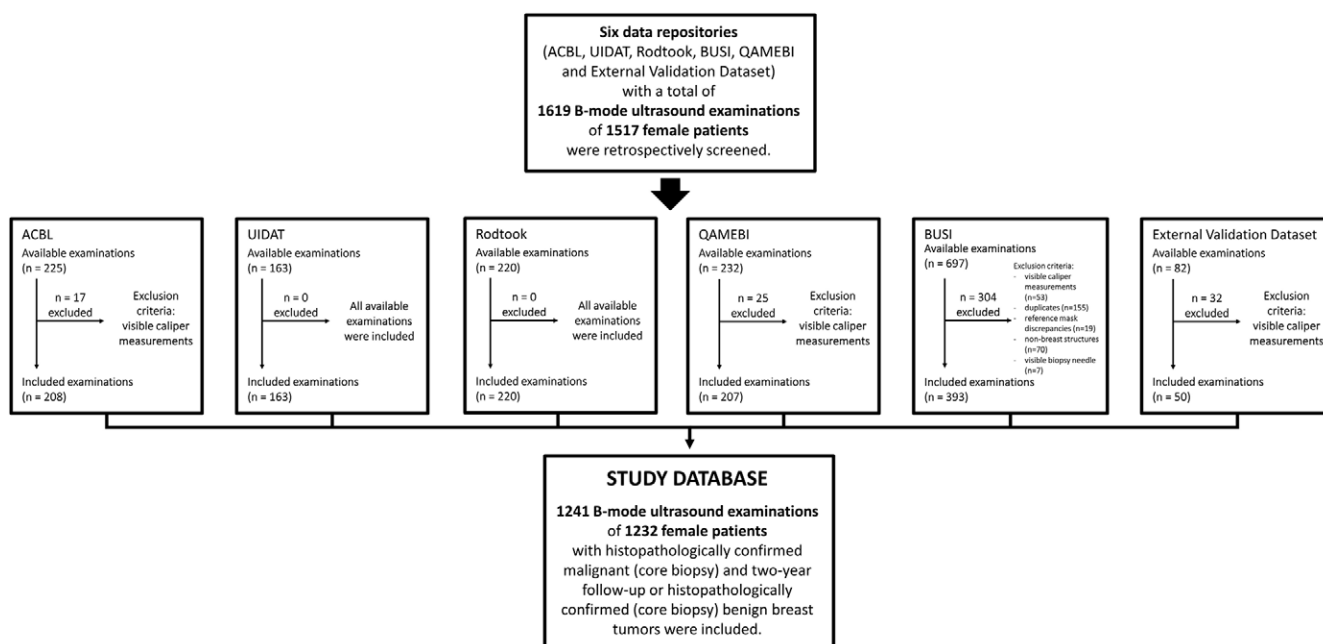


Figure 1: Flowchart of the inclusion and exclusion criteria of retrospectively reviewed 1619 B-mode US examinations in 1517 female patients. Exclusion criteria comprised examinations with visible calipers, duplicates, reference mask discrepancies, nonbreast structures, and visible biopsy needles. n = number of examinations.

captured the most representative features of malignant or benign lesions (17) (Appendix S2). Ground truth segments were manually delineated by two readers (M.K. and F.K.) with 3 and 20 years of experience in breast US imaging, respectively. Each reader performed the delineation once. In addition to the data acquired at the authors' institutions (Aachen Breast Lesions and external validation cohort), four public repositories were considered: UIDAT (18), Rodtook (19), BUSI (20), and QAMEBI (21) (Table 1).

The study database was randomly split at the examination level into two data sets with a uniform distribution of images with malignant and benign findings (Fig 2A). All data splits were processed in the same way. Data set 1 was used to develop breast lesion segmentation and was split into training and test sets with a 70:30 ratio. Data set 2 was used to develop breast lesion categorization and was split into feature search, train, and test set with a 50:35:15 ratio. In both instances, no validation set was formed as the training was performed with a fivefold cross-validation technique (22).

US Imaging

US imaging was performed (M.P., B.S.W., T.P., and K.K.) with Acuson Antares system (Siemens Healthineers) equipped with a 13.5-MHz transducer (VFX13–5), Aplio 500 scanner (Canon Medical Systems) with a 14-MHz transducer (PLT 1005BT), and Supersonic Imagine Aixplorer (SuperSonic Imagine) with a 15-MHz transducer (SL-15–4). After localization of breast tumors in B mode, shear-wave elastography measurement (USSW-A500A Shear Wave Kit; MI 1.5 ± 0.1) was performed through the largest tumor cross-section and the plane perpendicular to it. Additional Doppler US was performed.

US-based Radiomics

Two-dimensional nnU-Net (23) segmentation networks were trained for breast lesion segmentation. The trained models were evaluated regarding precision with Dice score (DS) and reproducibility (Fig 2B). A detailed description of our breast lesion segmentation framework is provided in Appendix S3.

The best segmentation model was used to localize the lesion candidates from data set 2. In the case of missed detection (DS = 0), the entire image was considered a "segment." The under-segmentations ($0 < DS < 0.5$) were included in the feature extraction process. Image-based features were computed with classic radiomics and convolutional autoencoder and extracted locally using tumor segments or bounding boxes and globally using the entire US image (Fig 2C). Additional information regarding our feature extraction and selection framework is provided in Appendix S4 and Figure S1.

Machine learning models were trained for breast tumor categorization using the selected radiomics signatures. A detailed description of the training procedure is provided in Appendix S5. Trained models were evaluated for AUC, sensitivity, specificity, positive predictive value, and negative predictive value. Moreover, the diagnostic performance of the highest-performing model was compared with that of an experienced reader (F.K.) and junior reader (M.K.).

Real-time Application

B-mode US videos of two patients with benign and malignant breast tumors were evaluated. The video frames were extracted and analyzed with the highest-performing breast tumor categorization model. Afterward, each frame that contained a lesion candidate was assigned the probability of malignant diagnosis, which was converted into a Breast Imaging Reporting and Data System (BI-RADS) score (24). The analyzed frames were reassembled into videos.

Statistical Analysis

To study reproducibility, the test sets of data set 1 and data set 2 were duplicated and speckle noise with variance of 0.05 was added to the images. Segments obtained from both the original and noisy images were evaluated by calculating the DS with reference to the ground truth (segments manually delineated by a radiologist). The equivalence test with null hypothesis that segmentation method is not reproducible was performed. Statistically significant result was set at $P < .05$.

Youden cutoff values and 95% CIs for AUC, sensitivity, specificity, positive predictive value, and negative predictive value were computed across 1000 bootstrap replications for all trained models. The two-tailed DeLong test (25) followed by Bonferroni test for multiple comparisons was used to compare the AUCs of derived models. Statistically significant difference was set at $P < .05$. Cohen κ coefficients (26) were calculated to measure the interreader reliability between the trained categorization model and the reference standard.

The sample size selection for the study was based on previous experience in developing artificial intelligence-based methods for segmentation and classification tasks. Groups were compared using either the independent two-tailed t test for continuous variables with normal distribution or the two-tailed Wilcoxon rank sum test for variables with nonnormal distribution. Statistically significant difference was considered at $P < .05$.

Results

Patients and Study Collective Characteristics

A total of 1619 B-mode US examinations of 1517 female patients were retrospectively screened. Of those, 1191 examinations of 1182 patients (mean age, 61 years $\pm$ 14 [SD]) were included in the development set and 50 examinations of 50 patients (mean age, 55 years $\pm$ 15) were included in the external validation data set (Table 1). Ninety-five examinations were excluded due to presence of visible caliper measurements, 155 due to duplicating breast tumor examinations, 19 due to discrepancies in the reference mask, 70 due to examinations showing structures other than breasts, and seven due to visible biopsy needles in the examination (Fig 1).

The development set was split at the examination level into two data sets: data set 1 (598 lesions) and data set 2 (593 lesions) with an equal distribution of benign and malignant cases (Table 2). To prevent data leakage, examinations of the same patient were included in only one data partition. Data set 1 was divided into a training set of 419 (70%) and test set of 179 (30%) lesions. Data set 2 was split into feature search set of 301

Table 1: Characteristic of the Study Database

Characteristic	ACBL	UIDAT	Rodtook	BUSI	QAMEBI	External Validation Data Set	<i>P</i> Value
No. of patients	208	163	215	389	207	50	...
Patient age (y)*	61 (14)	...	20–73	25–75	above 18	55 (15)	.42
No. of images	287	163	220	410	207	50	...
No. of lesions							
Overall	208	163	220	393	207	50	...
Benign	101	109	122	236	95	29	...
Malignant	107	54	98	157	112	21	...
Lesions size (mm)†							
Overall	18 (8)	...	...	...	...	16 (10)	.14
Benign	12 (7)	...	...	...	...	15 (8)	.49
Malignant	20 (8)	...	...	...	...	18 (12)	.17
BI-RADS range‡							
Benign	1–4	...	...	...	...	2–4	...
Malignant	4–6	...	...	...	...	4–6	...
Medical facility location	Germany	Spain	Thailand	Egypt	Iran	Germany	...
US scanner	ACUSON Antares (Siemens Healthineers); Aplio 500 (Toshiba/Canon)	ACUSON Sequoia C512 (Siemens Healthineers)	Philips iU22	LOGIQ E9; LOGIQ E9 Agile	AirPolar Ultimate	Supersonic Imagine Aixplorer	...
US transducer	13.5-MHz linear (VFX13–5); 14-MHz linear (PLT 1005BT)	8.5-MHz linear (17 L5 HD)	Not specified	5-MHz linear (ML6–15-D)	18-MHz linear	15-MHz linear (SL-15–4)	...
Ground truth							
Localization	Lesions delineated by a single experienced radiologist	Lesions delineated by a single experienced radiologist	Lesions delineated by three expert radiologists; final delineations obtained by majority voting	Lesions delineated by a single experienced radiologist	Lesions delineated by a single experienced radiologist	Lesions delineated by a single experienced radiologist	...
Characterization§	Histopathologically confirmed benign BI-RADS 4 and malignant BI-RADS 4–6; clinical follow-up–confirmed benign BI-RADS 2–3	Histopathologically confirmed malignant and benign	Histopathologically confirmed malignant and follow-up–confirmed benign	...	Histopathologically confirmed malignant and benign	Histopathologically confirmed benign BI-RADS 4 and malignant BI-RADS 4–6; histopathologically confirmed benign BI-RADS 2–3	...

Note.—Aachen Breast Lesions (ACBL) and external validation data sets were acquired at the authors' institution. UIDAT, Rodtook, BUSI, and QAMEBI are public repositories. Their use in this study was approved by the principal authors. Unless otherwise indicated, data are numbers of patients, images, and lesions in each data repository. BI-RADS = Breast Imaging Reporting and Data System.

* Data are means and data in parentheses are SDs. The principal authors of UIDAT (18) did not provide the age of patients; Rodtook (19) and BUSI (20) provided the range of patients' age; QAMEBI (21) recruited patients above 18 years old. *P* values for comparison of patients in ACBL and external validation dataset were calculated using the two-tailed Wilcoxon rank sum test for continuous variables with nonnormal distribution.

† Data are means and data in parentheses are SDs. The size of the lesions in UIDAT (18), Rodtook (19), BUSI (20), and QAMEBI (21) was not provided by the principal authors. *P* values for comparison of patients in ACBL and external validation dataset were calculated using the two-tailed Wilcoxon rank sum test for continuous variables with nonnormal distribution.

‡ Data in cells are BI-RADS ranges of included lesions. The BI-RADS range in UIDAT (18), Rodtook (19), BUSI (20), and QAMEBI (21) was not provided by the principal authors.

§ For ACBL, the malignant diagnosis of BI-RADS 4–6 and benign diagnosis of BI-RADS 4 lesions was confirmed with core-needle biopsy. The benign diagnosis of BI-RADS 2 and 3 lesions was confirmed if the results of follow-up US examinations of at least 24 months remained unchanged. For UIDAT (18), the malignant or benign diagnosis was obtained with core biopsy performed after US examination. For Rodtook (19), the malignant diagnosis was confirmed with core-needle biopsy. The benign diagnosis was confirmed if the results of follow-up US examinations of at least 24 months remained unchanged. For BUSI (20), the information was not provided by principal authors. For QAMEBI (21), the malignant or benign diagnosis was obtained with either fine-needle aspiration, core-needle biopsy, or open biopsy. For external validation data set, the malignant diagnosis of BI-RADS 4–6 and benign diagnosis of BI-RADS 4 lesions was confirmed with core-needle biopsy. The benign diagnosis of the BI-RADS 2 and 3 lesions was confirmed with core-needle biopsy.

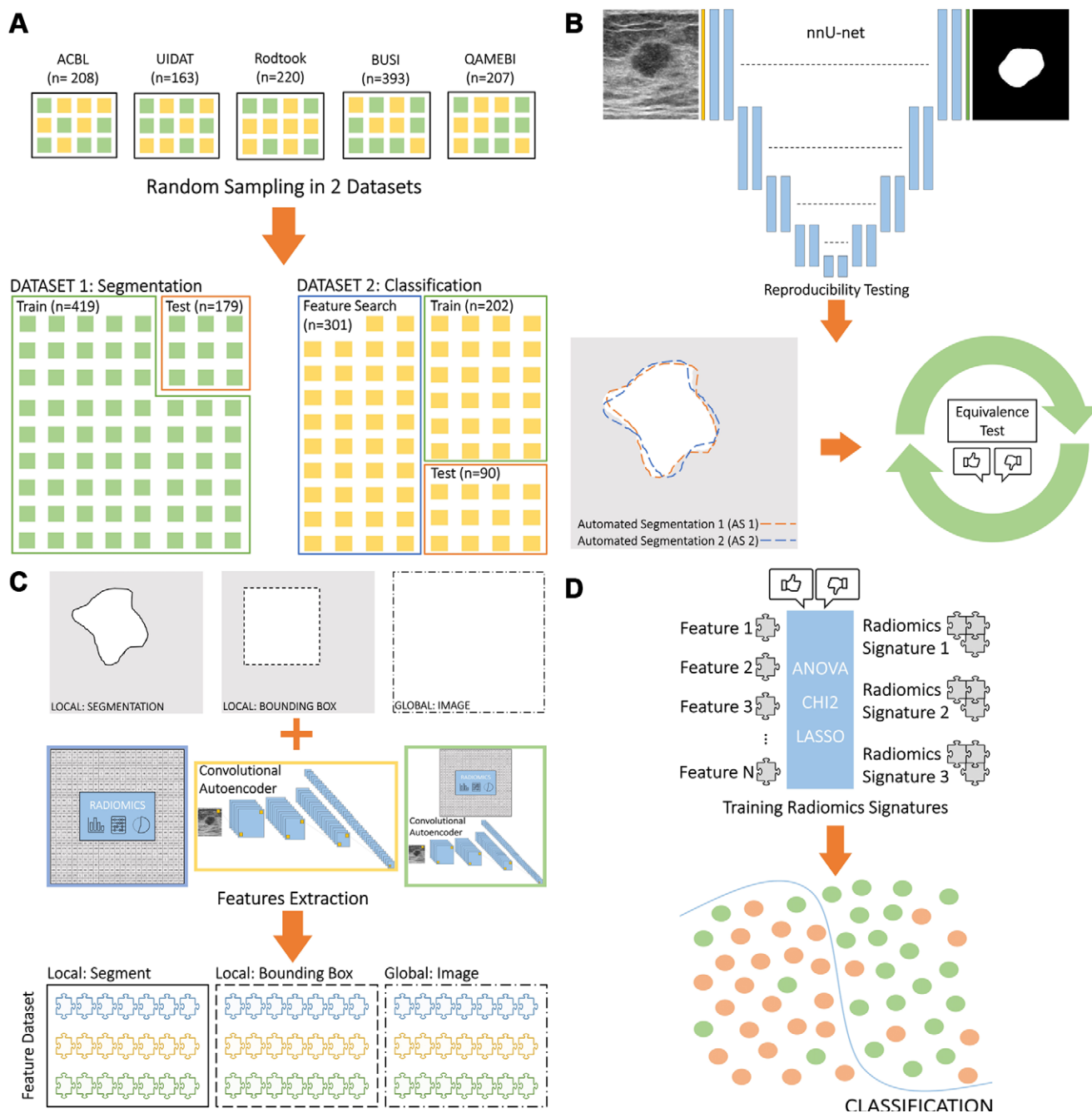


Figure 2: Overview of the study design. **(A)** B-mode US images from five independent data sets of female patients with breast tumors were randomly sampled into two data sets to develop breast lesion segmentation and classification frameworks. Aachen Breast Lesions (ACBL), UIDAT, Rodtook, BUSI, and QAMEBI are data repositories. **(B)** A nnU-Net (segmentation network) was trained for breast lesion segmentation. The reproducibility of the developed segmentation method was assessed with the equivalence test by statistically comparing the Dice scores calculated between the segments obtained on denoised (AS 1) and noisy (AS 2) images and the ground truth segments delineated by the experienced radiologist. **(C)** Lesion candidates were localized with tumor segments, tumor bounding boxes, and whole images. Image-based features were calculated with classic radiomics, convolutional autoencoder, and a mix of both approaches. The independent feature data sets—local segment, local bounding box, and global image—were assembled. Each feature data set comprised three groups of features: classic radiomics (blue), autoencoder (yellow), and mix (green). **(D)** The extracted features were subjected to stepwise feature selection to reveal the radiomics signatures—three per feature data set, which were later used as input for training breast cancer classification models.

(50%), train set of 202 (35%), and test set of 90 (15%) lesions. The feature search set was formed to eliminate over-fitting when training selected radiomics signatures.

US-based Radiomics

Performing a fivefold cross-validation training of nnU-Net resulted in six segmentation networks with different

configurations (nnU-Net-1 to 5 and nnU-Net-ensemble). The median DS for all segmentation networks was identical when applied to the test set of data set 1 at 0.90 and ranged from 0.88 to 0.89 when applied to data set 2 (Table 3, Fig 3). Equivalence testing showed that the segmentation approach is reproducible for segmentation and localization of breast lesions on B-mode US images ($P < .05$) (Table 3, Fig 4).

Table 2: Breast Lesion Segmentation and Categorization

Lesion type	Data Set 1		Data Set 2		
	Train	Test	Feature Search	Train	Test
Overall	419 (466)	179 (179)	301 (314)	202 (226)	90 (102)
Benign	254 (276)	108 (108)	151 (158)	105 (111)	45 (45)
Malignant	165 (190)	71 (71)	150 (156)	97 (115)	45 (57)

Note.—Each data set had uniform distribution of benign and malignant lesions. Data in parentheses are the total numbers of images included in the data split.

Table 3: Comparison between Tumor Segments Computed by nnU-Net Models and Physicians Using Median Dice Score

Characteristic	nnU-Net-1*	nnU-Net-2*	nnU-Net-3*	nnU-Net-4*	nnU-Net-5*	nnU-Net-ensemble*
Test: FN [†] 1 (0); FP [†] 19 (11)	0.90 (0.84–0.93)	0.90 (0.84–0.93)	0.90 (0.82–0.93)	0.90 (0.83–0.93)	0.90 (0.83–0.93)	0.90 (0.84–0.93)
<i>P</i> Value	.01	.01	.02	.01	.01	.02
Data set 2: FN [†] 23 (1); FP [†] 103 (63)	0.88 (0.80–0.92)	0.89 (0.80–0.92)	0.89 (0.80–0.92)	0.88 (0.80–0.92)	0.88 (0.80–0.92)	0.89 (0.80–0.92)
<i>P</i> Value	.001	.001	.001	.001	.001	.001

Note.—High agreement was observed between physicians and the automated segmentation models. *P* values for reproducibility testing of the segments generated by the automated segmentation model in denoised and noisy B-mode US images were calculated with the equivalence test. The values comprised in the table were computed with the use of the test set of data set 1 and all sets of data set 2. FN = false-negative, FP = false-positive.

* Data are Dice scores and data in parentheses are lower and upper bounds of IQR.

[†] Numbers outside parentheses represent either the total number of missed detections (FN) or total number of detections with Dice scores lower than 0.5 and higher than 0 (FP). Numbers in parentheses represent the total number of images that have been tagged with the same label (ie, FP or FN) by all nnU-Net models.

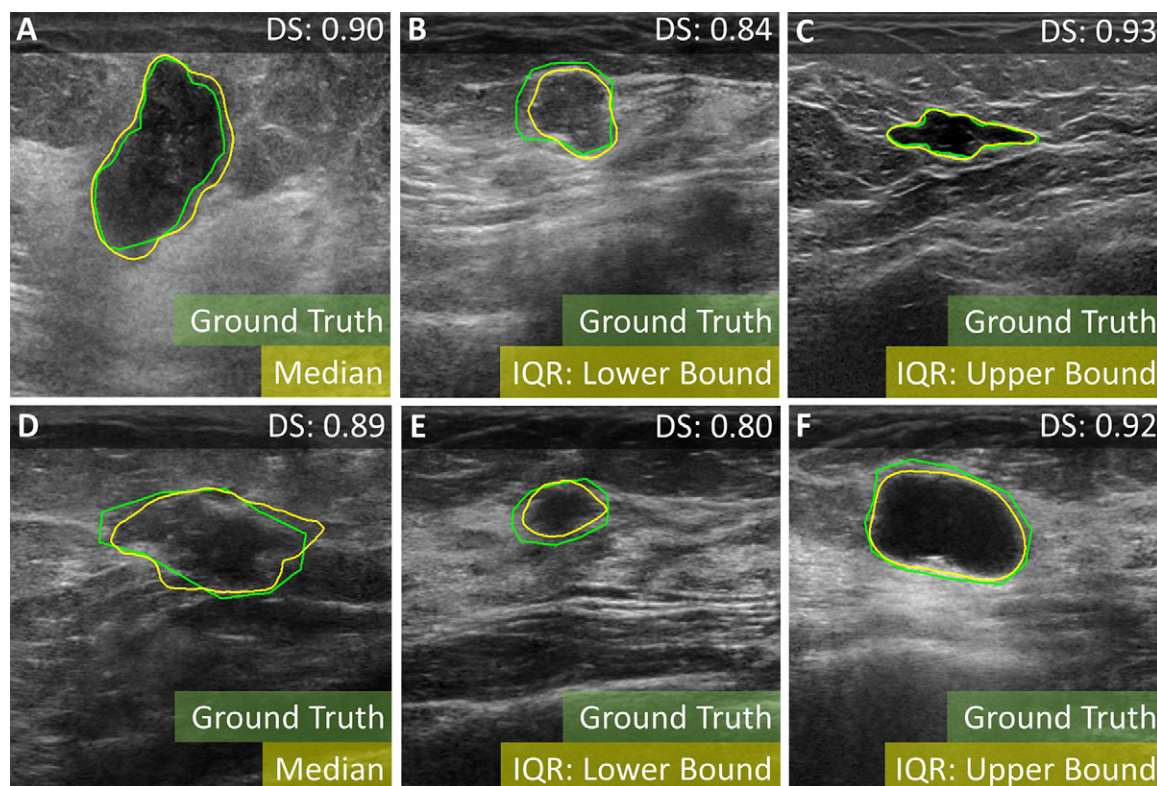


Figure 3: Examples of automated breast lesion segmentations on US images from (A–C) data set 1 and (D–F) data set 2 resulting in median, lower, and upper bound of IQR Dice scores (DS). The presented segments were computed by nnU-Net-2 (yellow) and experienced radiologist (green). The lesions were pathologically diagnosed as (A) malignant, (B) malignant, (C) malignant, (D) benign, (E) malignant, and (F) benign.

Selected Radiomics Signatures

Image-based features were extracted from five different regions of interest: (a) radiologist segmentation (local), (b) bounding box automatically assigned by the radiologist segmentation (local), (c) nnU-Net segmentation (local), (d) bounding box automatically assigned by nnU-Net segmentation (local), and (e) whole image (global). This resulted in five independent feature data sets. Each data set comprised three distinct groups of features: classic radiomics, autoencoder, and mixed. For classic radiomics, histogram-based, texture, and wavelet-based features were extracted. For autoencoder, features were extracted from the latent representation of the network after flattening the last layer of the encoder part.

Fifteen individual radiomics signatures were found. The median number of features per radiomics signature was 53 (IQR: 46, 68). The classic radiomics-based and mixed radiomics signatures mainly comprised textural and wavelet-based features. The autoencoder-based radiomics signatures comprised only of textural features. The summary of the composition of selected radiomics signatures is presented in Table S1.

Breast Tumor Categorization Results

Classic radiomics, autoencoder, and mixed radiomics signatures were used to categorize benign and malignant breast tumors on US images. Mixed features–based categorization models expressed the highest ability in discriminating benign and malignant breast tumors, with the AUC ranging from 0.72 to 0.90 (Table 4, Fig 5G). The model derived from radiologist bounding boxes obtained the highest AUC of 0.90 (95% CI: 0.83, 0.97) (Table 4). Comparison between the model and histopathologically or follow-up–confirmed diagnosis showed no evidence of a difference (AUC = 0.90 [95% CI: 0.83, 0.97] vs 1.00 [95% CI: 1.00, 1.00]; $P = .10$) (Table 4; Fig 5H, 5I). Accordingly, the radiologist bounding box–based model obtained the highest sensitivity of 81% (46 of 57; 95% CI: 70, 91) and specificity of 87% (39 of 45; 95% CI: 77, 87) (Table 4). The positive and negative predictive values of this model were 88% (46 of 52; 95% CI: 80, 97) and 78% (39 of 50; 95% CI: 67, 89), respectively. The second-best model was derived from nnU-Net bounding boxes (Table 4). Comparison between this model and histopathologically or follow-up–confirmed breast tumor diagnosis was statistically different (AUC = 0.85 [95% CI: 0.77, 0.93] vs 1.00 [95% CI: 1.00, 1.00]; $P = .02$) (Table 4; Fig 5H, 5I). For all three feature groups, the breast tumor categorizations derived from either radiologist or nnU-Net bounding boxes achieved the highest diagnostic performance (Table 4). Inclusion of multiple

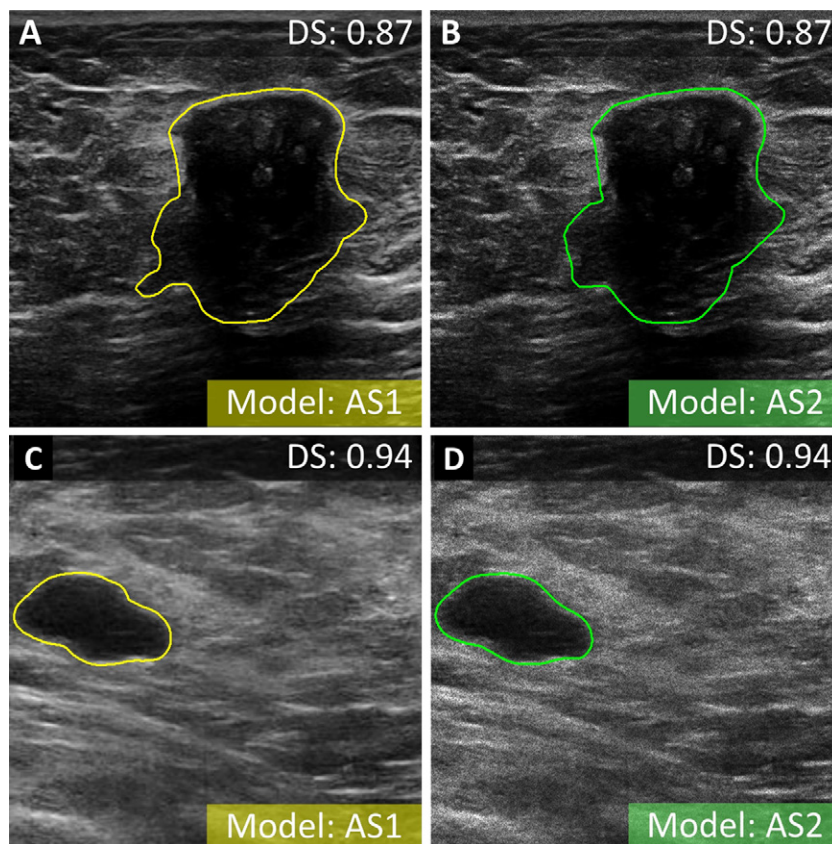


Figure 4: Representative examples of automated breast lesion segmentations on US images from test sets of (A, B) data set 1 and (C, D) data set 2. The lesions were pathologically diagnosed as (A, B) malignant and (C, D) benign. The automated segmentations (AS) were performed twice (AS1 and AS2): first in original images (A, C) and then in the same images with added speckle noise (B, D).

examinations from the same patient in a single data split had no effect on the breast lesion categorization ($P > .05$).

Comparison of Diagnostic Performance between Model and Readers

The AUC of the experienced and junior US reader for the test set was 0.83 (95% CI: 0.76, 0.90) and 0.82 (95% CI: 0.75, 0.90), respectively (Fig 6A). The model obtained an AUC of 0.90 (95% CI: 0.83, 0.97) for the same data set. There was no evidence of a difference between the AUCs of the model and the readers (AUC = 0.90 [95% CI: 0.83, 0.97] vs 0.83 [95% CI: 0.76, 0.90]; $P = .55$ and AUC = 0.90 [95% CI: 0.83, 0.97] vs 0.82 [95% CI: 0.75, 0.90]; $P = .45$) (Fig 6B). However, the model differed from human readers in the subjective categorization of breast lesions (κ values = 0.15 and 0.24) (Fig 6C). In the external validation data set, although the AUC of the model (0.70; 95% CI: 0.55, 0.84) was lower than that of the experienced (AUC = 0.72; 95% CI: 0.62, 0.83) and junior (AUC = 0.73; 95% CI: 0.62, 0.83) readers, there was no evidence of a difference between them (AUC = 0.70 [95% CI: 0.55, 0.84] vs 0.72 [95% CI: 0.62, 0.83]; $P = .58$ and AUC = 0.70 [95% CI: 0.55, 0.84] vs 0.73 [95% CI: 0.62, 0.83]; $P = .11$) (Fig 6D, 6E). However, again the model differed from both human readers in the subjective categorization of breast lesions (κ values = 0.30 and 0.46) (Fig 6F).

Table 4: Diagnostic Performance of Breast Cancer Classification Models Built on Classic Radiomics, Autoencoder, or Mixed Features Derived from Whole Image, Radiologist Segmentation, Radiologist Bounding Box, nnU-Net Segmentation, and nnU-Net Bounding Box

Variable	Classic Radiomics	<i>P</i> Value*	Autoencoder	<i>P</i> Value*	Mixed	<i>P</i> Value*
AUC						
Whole image	0.64 [0.53, 0.75]	.00	0.63 [0.52, 0.74]	.00	0.72 [0.62, 0.82]	.00
Radiologist segmentation	0.79 [0.70, 0.88]	.02	0.64 [0.53, 0.75]	.01	0.83 [0.75, 0.91]	.02
Radiologist bounding box	0.82 [0.73, 0.91]	.04	0.80 [0.71, 0.89]	.05	0.90 [0.83, 0.97]	.10
nnU-Net segmentation	0.78 [0.69, 0.87]	.01	0.70 [0.60, 0.80]	.00	0.78 [0.69, 0.87]	.07
nnU-Net bounding box	0.83 [0.75, 0.91]	.03	0.82 [0.73, 0.91]	.01	0.85 [0.77, 0.93]	.02
Sensitivity (%)						
Whole image	47 (27/57) [34, 60]	...	49 (28/57) [36, 62]	...	60 (34/57) [47, 72]	...
Radiologist segmentation	74 (42/57) [62, 85]	...	77 (44/57) [66, 88]	...	70 (40/57) [58, 82]	...
Radiologist bounding box	74 (42/57) [62, 85]	...	81 (46/57) [70, 91]	...	81 (46/57) [70, 91]	...
nnU-Net segmentation	75 (43/57) [64, 87]	...	51 (29/57) [38, 64]	...	88 (50/57) [79, 96]	...
nnU-Net bounding box	77 (44/57) [66, 88]	...	75 (43/57) [64, 87]	...	70 (40/57) [58, 82]	...
Specificity (%)						
Whole image	60 (27/45) [46, 60]	...	67 (30/45) [53, 67]	...	67 (30/45) [53, 67]	...
Radiologist segmentation	62 (28/45) [48, 62]	...	49 (22/45) [34, 49]	...	78 (35/45) [66, 78]	...
Radiologist bounding box	71 (32/45) [58, 71]	...	62 (28/45) [48, 62]	...	87 (39/45) [77, 87]	...
nnU-Net segmentation	69 (31/45) [55, 69]	...	76 (34/45) [63, 76]	...	44 (20/45) [30, 44]	...
nnU-Net bounding box	71 (32/45) [58, 71]	...	84 (38/45) [74, 84]	...	80 (36/45) [68, 80]	...
PPV (%)						
Whole image	60 (27/45) [46, 74]	...	65 (28/43) [51, 79]	...	69 (34/49) [56, 82]	...
Radiologist Segmentation	71 (42/59) [60, 83]	...	66 (44/67) [54, 77]	...	80 (40/50) [69, 91]	...
Radiologist Bounding Box	76 (42/55) [65, 88]	...	73 (46/63) [62, 84]	...	88 (46/52) [80, 97]	...
nnU-Net segmentation	75 (43/57) [64, 87]	...	73 (29/40) [59, 86]	...	67 (50/75) [56, 77]	...
nnU-Net bounding box	77 (44/57) [66, 88]	...	86 (43/50) [76, 96]	...	82 (40/49) [71, 92]	...
NPV (%)						
Whole image	47 (27/57) [34, 60]	...	51 (30/59) [38, 64]	...	57 (30/53) [43, 70]	...
Radiologist segmentation	65 (28/43) [51, 79]	...	63 (22/35) [47, 79]	...	67 (35/52) [55, 80]	...
Radiologist bounding box	68 (32/47) [55, 81]	...	72 (28/39) [58, 86]	...	78 (39/50) [67, 89]	...
nnU-Net segmentation	69 (31/45) [55, 82]	...	55 (34/62) [42, 67]	...	74 (20/27) [58, 91]	...
nnU-Net bounding box	71 (32/45) [58, 71]	...	73 (38/52) [61, 85]	...	68 (36/53) [55, 80]	...

Note.—The models with the highest area under the receiver operating characteristic (AUC) were the model derived from the radiologist's bounding box, followed by the model using the nnU-net bounding boxes. Data in parentheses are numerator/denominator and represent the number of images in the test set; data in brackets are 95% CIs. NPV = negative predictive value, PPV = positive predictive value.

* *P* values are for the comparison of the AUC of the models with the AUC of histopathologically or follow-up–confirmed diagnosis of breast tumor (AUC = 1.00 [95% CI: 1.00, 1.00]) and were calculated with the DeLong test. All *P* values were Bonferroni multihypothesis corrected.

Potential Real-Time Implementation

The model with the highest breast tumor categorization ability (AUC = 0.90 [95% CI: 0.83, 0.97]) (Table 4) was applied to two B-mode US cine clips of a malignant (BI-RADS 5) and a benign (BI-RADS 2) lesion. Since Orel et al (24) found that 23% of BI-RADS 4 lesions are malignant (calculated by relating BI-RADS findings to the outcome of biopsies performed), we defined a lesion as BI-RADS 4 and higher if the decision support system assigned a probability of malignancy of 23% or more. If, on the other hand, the probability of a malignant diagnosis was less than 23%, we defined the lesion as BI-RADS 3 and lower. The demonstration of the real-time implementation of our model is available in Movies 1 and 2 and Figure S2.

Discussion

In the retrospective study, we developed an effective real-time lesion localization and categorization method that could help overcome the user-dependency of breast US imaging.

nnU-Net–based breast lesion localization proved precision and reproducibility in test set of data set 1 (median DS: 0.90 [IQR, 0.84–0.93]; *P* = .01) and in data set 2 (median DS: 0.89 [IQR, 0.80–0.92]; *P* = .001). Combining radiomics and auto-encoder features derived from tumor bounding boxes facilitated the development of real-time breast tumor categorization that achieved AUC of 0.90 (95% CI: 0.83, 0.97), sensitivity of 81% (46 of 57; 95% CI: 70, 91), and specificity of 87% (39 of 45; 95% CI: 77, 87). No difference was found between the model and histopathologically confirmed breast tumor diagnoses (AUC

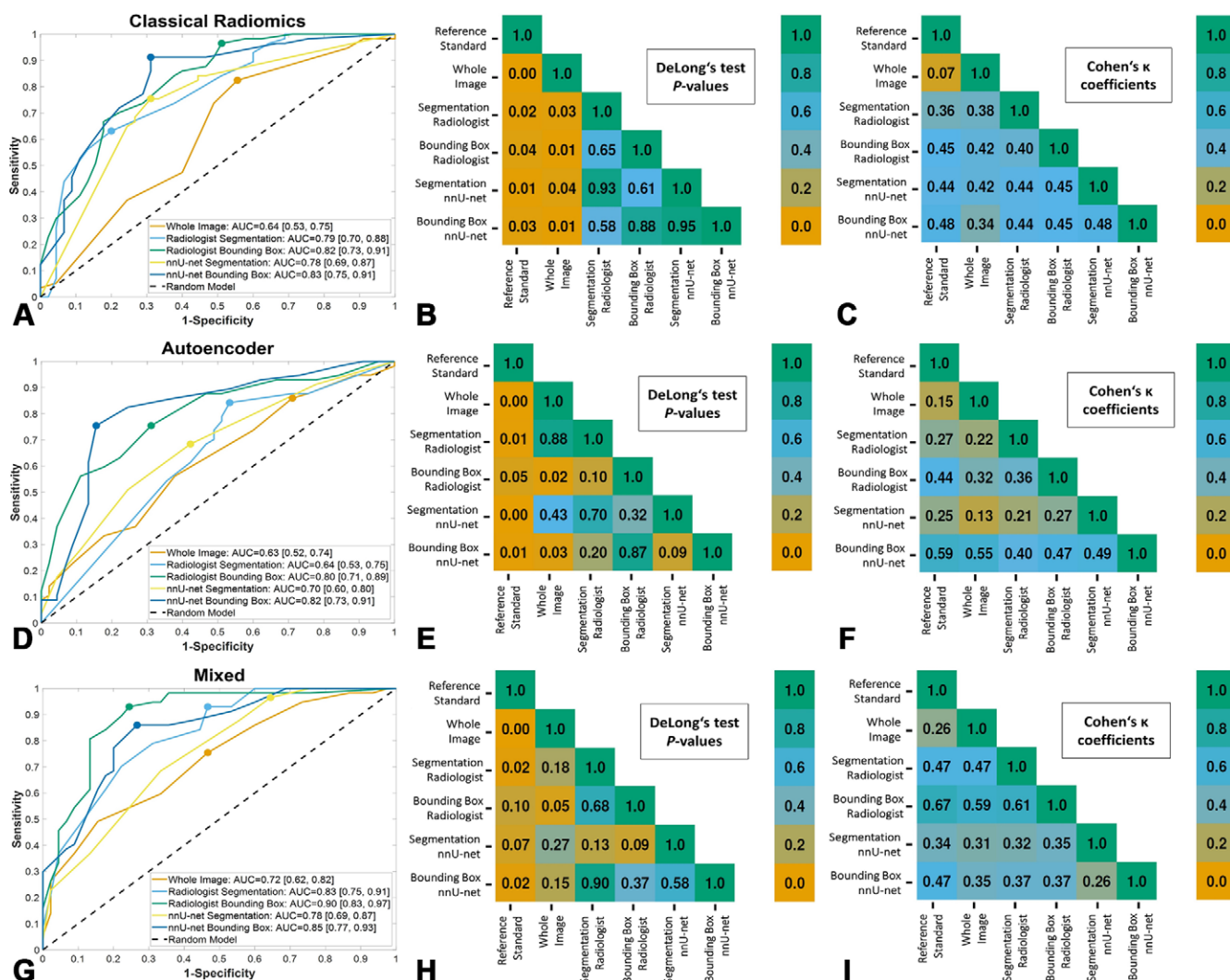


Figure 5: Breast cancer classification study results. (A) Area under the receiver operating characteristic curves (AUCs) for breast cancer classification models trained on classic radiomics features. The diagnostic performance of the model derived from nnU-Net bounding boxes exceeds the other models. (B) Heat map shows the statistical comparison, measured with two-tailed DeLong test (23). The difference between models and reference standard was significant ($P < .05$). (C) Heat map shows the agreement, measured with Cohen κ (24). The models are in slight ($\kappa = 0.07$) to moderate ($\kappa = 0.48$) agreement with reference standard. (D) AUCs for breast cancer classification models trained on autoencoder features. The diagnostic performance of the model derived from nnU-Net bounding boxes exceeds the other models. (E) Heat map shows the statistical comparison, measured with two-tailed DeLong test (23). There was no difference between the model derived from radiologist bounding boxes and reference standard ($P > .05$). (F) Heat map shows the agreement, measured with Cohen κ (24). The models are in slight ($\kappa = 0.15$) to substantial ($\kappa = 0.59$) agreement with reference standard. (G) AUCs for breast cancer classification models trained on mixed features. The diagnostic performance of the model derived from radiologist bounding boxes exceeds the other models. (H) Heat map shows the statistical comparison, measured with two-tailed DeLong test (23). There was no difference between the models derived from radiologist bounding boxes ($P = .10$) or nnU-Net segments ($P = .07$) and reference standard. (I) Heat map shows the agreement, measured with Cohen κ (24). The models are in moderate ($\kappa = 0.26$) to substantial ($\kappa = 0.67$) agreement with the reference standard. The reference standard is histopathologically or follow-up-confirmed breast tumor diagnosis. Data in brackets are 95% CIs. All P values were Bonferroni multihypothesis corrected. The points on the curves mark the Youden cutoff point of each AUC.

= 0.90 [95% CI: 0.83, 0.97] vs 1.00 [95% CI: 1.00, 1.00]; $P = .10$) nor in breast lesion categorization between the model and human readers (AUC = 0.90 [95% CI: 0.83, 0.97] vs 0.83 [95% CI: 0.76, 0.90]; $P = .55$ and 0.90 [95% CI: 0.83, 0.97] vs 0.82 [95% CI: 0.75, 0.90]; $P = .45$).

Mixed-features models performed best, possibly due to their capacity to capture the textural patterns at low (classic radiomics) and high (autoencoders) levels of image abstraction (7,15). Previous findings showed that combining decisions from two models trained on different features improves the diagnostic accuracy of US-based tumor categorization (14,27). Unlike these studies, which use two different models, our method notably simplifies

categorization by eliminating the need for multiple algorithms and training.

This study is, to the best of our knowledge, the first to use nnU-Net for breast lesion localization on B-mode US images. We showed that our lesion localization is reproducible, independent of added noise ($P < .01$, $P < .001$). Moreover, the use of nnU-Net reduced the false-negative rate from 15% to 1% compared with our previous study (16), substantially increasing the robustness of our categorization approach.

Unlike detailed segments, tumor bounding boxes encompass adjacent breast tissue, which is known to undergo significant remodeling at the in situ carcinoma stage (28) and further

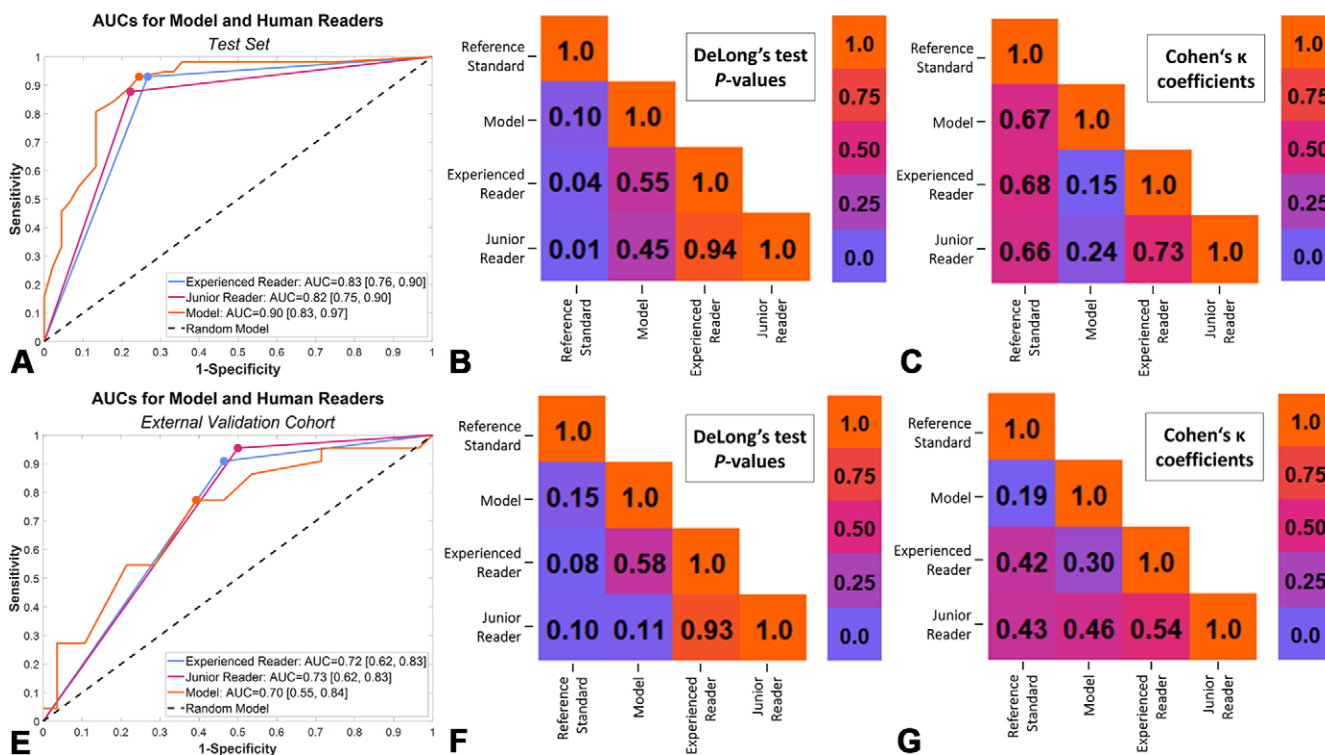


Figure 6: Comparison of diagnostic performance between model and human readers. (A) Area under the receiver operating characteristic curves (AUCs) of experienced and junior readers and model derived on the test set. The model outperformed readers in both sensitivity and specificity. (B) Heat map shows the statistical comparison, measured with two-tailed DeLong test (23). There is no difference between model and reference standard ($P = .10$) and model and human readers (experienced reader: $P = .55$; junior reader: $P = .45$). (C) Heat map shows the agreement, measured with Cohen κ (24). Both the model and readers agree with the reference standard (κ values between 0.66 and 0.68). The model differs from human readers (κ values between 0.15 and 0.24). (D) AUCs of experienced and junior readers and model derived on the external validation cohort. The model outperforms readers in the high specificity range. (E) Heat map shows the statistical comparison, measured with two-tailed DeLong test (23). There is no difference between both model and reference standard ($P = .15$) and model and human readers (experienced reader: $P = .58$; junior reader: $P = .11$). (F) Heat map represents the agreement, measured with Cohen κ (24). The model is in fair agreement ($\kappa = 0.19$), while both human readers ($\kappa = 0.42$ and 0.43) are in moderate agreement with reference standard. The model is in good agreement with both human readers ($\kappa = 0.30$ and 0.46). The reference standard is histopathologically or follow-up confirmed breast tumor diagnosis. Data in brackets are 95% CIs. The points on the curves mark the optimal operating point of each AUC.

microenvironmental responses during tumor invasion and progression. Nevertheless, deriving image-based features from detailed tumor segments is still the most popular method (29,30), and only several studies showed that segmentation-free breast tumor categorization could exhibit higher discriminative ability (11,15). We compared the diagnostic performance of breast tumor categorization models derived from tumor bounding boxes, detailed tumor segmentation obtained either by an experienced radiologist (ground truth) or by a state-of-the-art segmentation network (nnU-Net) and from whole US images. We showed that bounding box-based categorization can better discriminate between malignant and benign breast lesions, independently of the used image-based features. These results not only match but exceed those of our previous study (16).

Our primary motivation is to directly integrate our model into clinical US systems. Previous studies (31,32) showed that breast US examination without computer-assisted tumor diagnosis has overall sensitivity of 67% and specificity of 77%. Our model surpassed these values, achieving 81% (46 of 57) sensitivity and 87% (39 of 45) specificity. Furthermore, our framework can score malignancy of breast lesion detected in real time. Introducing this advancement to standard breast US imaging could serve as a second opinion for both experienced and

inexperienced operators. Previously, it was shown that mainly inexperienced operators gain the most benefit from computer-assisted breast tumor diagnosis, especially in terms of sensitivity improvement (33). However, it could also increase the specificity of US imaging performed by experienced readers, thereby reducing the number of ordered biopsies (31,32). We found no statistical difference between model, experienced reader, or junior reader ($P = .55$; $P = .45$). Therefore, we believe that our framework could become reliable assistance for standard US-based breast examination performed by any users. It also might have an impact on reducing intraobserver variability.

Our study had several limitations. First, we performed the analysis on US images representing a single cross-section that included the most suspicious features of breast lesions. This might influence the specificity of the diagnosis made by our model in real time. Using the full stack of US images may improve the robustness of our breast lesion categorization method. Second, the ground truth in our database was obtained by one radiologist for each included data set. Because we included six different data sets, the risk of potential bias is reduced. Nevertheless, for future studies, our plan is to refine our methodology to avoid relying on predetermined labels. Third, using a localization method relying solely on bounding boxes, for example, region-based

convolutional neural networks (34), may lower the false-positive rate and substantially increase the detection speed, thus, improving the real-time applicability of our breast tumor categorization model. Finally, our breast tumor categorization achieved sensitivity of 81% (46 of 57) and specificity of 87% (39 of 45) thus far. We aim to further improve this by adaptively optimizing the decision threshold of our breast tumor categorization, refining the feature extraction methods and composition of selected radiomics signatures to tailor our tool for clinical use.

In conclusion, we developed an effective US-based breast tumor categorization model derived from tumor bounding boxes with a mix of classic radiomics and autoencoder-based features. The model may be used alone or together with the reference standard for categorization of benign or malignant breast tumors. Our model has the potential to assist physicians in breast examination during and after US imaging and we will now focus our research on improving the real-time approach.

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References

- Gilbert FJ, Pinker-Domenig K. IDKD Springer Series Diagnosis and Staging of Breast Cancer: When and How to Use Mammography, Tomosynthesis, Ultrasound, Contrast-Enhanced Mammography, and Magnetic Resonance Imaging. In: Hodler J, Kubik-Huch RA, von Schulthess GK, eds. Diseases of the Chest, Breast, Heart and Vessels 2019-2022: Diagnostic and Interventional Imaging. Springer, 2019; 155–166.
- Mann RM, Athanasiou A, Baltzer PAT, et al. Breast cancer screening in women with extremely dense breasts recommendations of the European Society of Breast Imaging (EUSOBI). *Eur Radiol* 2022;32(6):4036–4045.
- Mann RM, Hooley R, Barr RG, Moy LJR. Novel approaches to screening for breast cancer. *Radiology* 2020;297(2):266–285.
- Gillies RJ, Schabath MB. Radiomics Improves Cancer Screening and Early Detection. *Cancer Epidemiol Biomarkers Prev* 2020;29(12):2556–2567.
- Romeo V, Cuocolo R, Apolito R, et al. Clinical value of radiomics and machine learning in breast ultrasound: a multicenter study for differential diagnosis of benign and malignant lesions. *Eur Radiol* 2021;31(12):9511–9519.
- Fleury E, Marcomini K. Performance of machine learning software to classify breast lesions using BI-RADS radiomic features on ultrasound images. *Eur Radiol Exp* 2019;3(1):34.
- Gillies RJ, Kinahan PE, Hricak H. Radiomics: Images Are More than Pictures, They Are Data. *Radiology* 2016;278(2):563–577.
- O'Connor JPB, Aboagye EO, Adams JE, et al. Imaging biomarker roadmap for cancer studies. *Nat Rev Clin Oncol* 2017;14(3):169–186.
- Luo WQ, Huang QX, Huang XW, Hu HT, Zeng FQ, Wang W. Predicting Breast Cancer in Breast Imaging Reporting and Data System (BI-RADS) Ultrasound Category 4 or 5 Lesions: A Nomogram Combining Radiomics and BI-RADS. *Sci Rep* 2019;9(1):11921.
- Wang J, Kato F, Oyama-Manabe N, et al. Identifying Triple-Negative Breast Cancer Using Background Parenchymal Enhancement Heterogeneity on Dynamic Contrast-Enhanced MRI: A Pilot Radiomics Study. *PLoS One* 2015;10(11):e0143308.
- Zhou Y, Xu J, Liu Q, et al. A Radiomics Approach With CNN for Shear-Wave Elastography Breast Tumor Classification. *IEEE Trans Biomed Eng* 2018;65(9):1935–1942.
- Alvarenga AV, Infantosi AFC, Pereira WC, Azevedo CM. Assessing the combined performance of texture and morphological parameters in distinguishing breast tumors in ultrasound images. *Med Phys* 2012;39(12):7350–7358.
- Wu T, Sultan LR, Tian J, Cary TW, Sehgal CM. Machine learning for diagnostic ultrasound of triple-negative breast cancer. *Breast Cancer Res Treat* 2019;173(2):365–373.
- Antropova N, Huynh BQ, Giger ML. A deep feature fusion methodology for breast cancer diagnosis demonstrated on three imaging modality datasets. *Med Phys* 2017;44(10):5162–5171.
- Xu Z, Wang Y, Chen M, Zhang Q. Multi-region radiomics for artificially intelligent diagnosis of breast cancer using multimodal ultrasound. *Comput Biol Med* 2022;149:105920.
- Magnuska ZA, Theek B, Darguzyte M, et al. Influence of the Computer-Aided Decision Support System Design on Ultrasound-Based Breast Cancer Classification. *Cancers (Basel)* 2022;14(2):277.
- D'Orsi C, Bassett L, Feig S, et al. Breast imaging reporting and data system (BI-RADS). 4th ed. Reston, Va: American College of Radiology, 2018.
- Yap MH, Pons G, Marti J, et al. Automated Breast Ultrasound Lesions Detection Using Convolutional Neural Networks. *IEEE J Biomed Health Inform* 2018;22(4):1218–1226.
- Rodtook A, Kirimasthong K, Lohitvisate W, Makhanov SS. Automatic initialization of active contours and level set method in ultrasound images of breast abnormalities. *Pattern Recognit* 2018;79:172–182.
- Al-Dhabyani W, Gomaa M, Khaled H, Fahmy A. Dataset of breast ultrasound images. *Data Brief* 2019;28:104863.
- Abbasian Ardakani A, Mohammadi A, Mirza-Aghazadeh-Attari M, Acharya UR. An open-access breast lesion ultrasound image database: Applicable in artificial intelligence studies. *Comput Biol Med* 2023;152:106438.
- Soper DS. Greed is good: Rapid hyperparameter optimization and model selection using greedy k-fold cross validation. *Electronics (Basel)* 2021;10(16):1973.
- Isensee F, Jaeger PF, Kohl SAA, Petersen J, Maier-Hein KH. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nat Methods* 2021;18(2):203–211.
- Orel SG, Kay N, Reynolds C, Sullivan DC. BI-RADS categorization as a predictor of malignancy. *Radiology* 1999;211(3):845–850.
- DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics* 1988;44(3):837–845.
- Cohen J. A Coefficient of Agreement for Nominal Scales. *Educ Psychol Meas* 1960;20(1):37–46.
- Bove S, Comes MC, Lorusso V, et al. A ultrasound-based radiomic approach to predict the nodal status in clinically negative breast cancer patients. *Sci Rep* 2022;12(1):7914.
- Toss MS, Miligy IM, Gorringer KL, et al. Geometric characteristics of collagen have independent prognostic significance in breast ductal carcinoma in situ: an image analysis study. *Mod Pathol* 2019;32(10):1473–1485.
- Fleury EFC, Marcomini K. Impact of radiomics on the breast ultrasound radiologist's clinical practice: From lumpologist to data wrangler. *Eur J Radiol* 2020;131:109197.
- Shen Y, Shamout FE, Oliver JR, et al. Artificial intelligence system reduces false-positive findings in the interpretation of breast ultrasound exams. *Nat Commun* 2021;12(1):5645.
- Geisel J, Raghu M, Hooley R. The Role of Ultrasound in Breast Cancer Screening: The Case for and Against Ultrasound. *Semin Ultrasound CT MR* 2018;39(1):25–34.
- Aristokli N, Polycarpou I, Themistocleous SC, Sophocleous D, Mamais I. Comparison of the diagnostic performance of Magnetic Resonance Imaging (MRI), ultrasound and mammography for detection of breast cancer based on tumor type, breast density and patient's history: A review. *Radiography* 2022;28(3):848–856.
- Chabi ML, Borget I, Ardiles R, et al. Evaluation of the accuracy of a computer-aided diagnosis (CAD) system in breast ultrasound according to the radiologist's experience. *Acad Radiol* 2012;19(3):311–319.
- Wang CY, Bochkovskiy A, Liao HYM. YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors. In: 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2023; 7464–7475.